56) Prove that if x is a real number, then $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

Soln: Let, $\lfloor -x \rfloor = n$. By Defn. 12, $n \in \mathbb{Z}$.

By Prop 1(a), $n \leq -x < n+1$. $\therefore -n \geq x > -n-1 \Rightarrow -n-1 < x \leq -n$.

By Prob 1(b), $\lceil x \rceil = -n \Rightarrow n = -\lceil x \rceil \Rightarrow \lfloor -x \rfloor = -\lceil x \rceil$

Let, $\lceil -x \rceil = n$. By Defn. 12, $n \in \mathbb{Z}$.

By Prop 1(b), $n-1 < -x \leq n \Rightarrow -n+1 > x \geq -n \Rightarrow -n \leq x < -n+1$

By Prop 1(a), $\lfloor x \rfloor = -n \Rightarrow n = -\lfloor x \rfloor \Rightarrow \lceil -x \rceil = -\lfloor x \rfloor$ □

57) The function INT is found on some calculators, where $\text{INT}(x) = \lfloor x \rfloor$ where x is a nonnegative real number and $\text{INT}(x) = \lceil x \rceil$ when x is a negative real number. Show that this INT function satisfies the identity $\text{INT}(-x) = -\text{INT}(x)$.

Soln: Given, $x \in \mathbb{R}$. if $x \geq 0$, then $\text{INT}(x) = \lfloor x \rfloor$. if $x < 0$, then $\text{INT}(x) = \lceil x \rceil$

Let, $x \geq 0$. $\therefore -x \leq 0$. $\text{INT}(-x) = \begin{cases} \lceil -x \rceil \text{ if } -x < 0 \\ \lfloor 0 \rfloor \text{ if } x = 0 \end{cases} = \begin{cases} -\lfloor x \rfloor \text{ if } -x < 0 \\ 0 \text{ if } x = 0 \end{cases}$ (By Prob 56)

$= \begin{cases} -\lfloor x \rfloor \text{ if } -x < 0 \\ -0 \text{ if } x = 0 \end{cases} = \begin{cases} -\lfloor x \rfloor \text{ if } x > 0 \\ -0 \text{ if } x = 0 \end{cases}$

$= -\text{INT}(x)$

Let, $x < 0$. $\therefore -x > 0$. $\text{INT}(-x) = \lfloor -x \rfloor = -\lceil x \rceil$ (By Prob 56)

$= -\text{INT}(x)$.

$\therefore \text{INT}(x) = -\text{INT}(x)$ □

59) Let a and b be real numbers with $a < b$. Use the floor and/or ceiling functions to express the number of integers n that satisfy the inequality $a < n < b$.

Soln: By Property 2, $\lfloor a \rfloor + 1 > a$. and $b \leq \lceil b \rceil$. $\therefore \lfloor a \rfloor + 1 > a < n < b \leq \lceil b \rceil$.

$\therefore$ The number of integers n s.t. $a < n < b$ is $\lceil b \rceil - \lfloor a \rfloor - 1$ □

71) Find the inverse function of $f(x) = x^3 + 1$.

Soln: Let, $f(x) = y$ $\therefore f^{-1}(y) = x$. (By Defn. 9)

$\therefore x^3 + 1 = y \Rightarrow x = \sqrt[3]{y-1}$ $\quad f^{-1}(y) = \sqrt[3]{y-1}$ □